

Sales for Retail and Food Services in United States

1) Top-performing industries in terms of sales for a year 2021, and how do their sales compare month-over-month?

WITH monthly_sales AS (

SELECT

year,

month,

industry,

SUM(sales) AS total_sales

FROM

retail_sales

WHERE

year = 2021

GROUP BY

year,

month,

industry

),

top_industries AS (

SELECT

year,

month,

industry,

total_sales,

RANK() OVER (PARTITION BY year, month ORDER BY total_sales DESC) AS

industry_rank

FROM

monthly_sales

)

```

SELECT
    year,
    month,
    industry,
    total_sales
FROM
    top_industries
WHERE
    industry_rank = 1
ORDER BY
    year,
    month;

```

Data Output					Messages	Notifications
	year integer	month integer	industry text	total_sales bigint		
1	2021	1	Automotive	375492		
2	2021	2	Automotive	369166		
3	2021	3	Automotive	524244		
4	2021	4	Automotive	512810		
5	2021	5	Automotive	508814		
6	2021	6	Automotive	486902		
7	2021	7	Automotive	476584		
8	2021	8	Automotive	455964		
9	2021	9	Automotive	438496		
10	2021	10	Automotive	443780		
11	2021	11	Automotive	428848		
12	2021	12	Automotive	454888		
Total rows: 12 of 12					Query complete 00:00:00.079	

2)Top-performing industries in terms of sales for a year 2022, and how do their sales compare month-over-month?

WITH monthly_sales AS (

```

SELECT
    year,
    month,
    industry,
    SUM(sales) AS total_sales
FROM
    retail_sales
WHERE
    year = 2022
GROUP BY
    year,
    month,
    industry
),
top_industries AS (
    SELECT
        year,
        month,
        industry,
        total_sales,
        RANK() OVER (PARTITION BY year, month ORDER BY total_sales DESC) AS
industry_rank
    FROM
        monthly_sales
)
SELECT
    year,
    month,
    industry,
    total_sales
FROM
    top_industries
WHERE
    industry_rank = 1

```

ORDER BY

year,

month;

Data Output		Messages		Notifications					
									
	year integer		month integer		industry text		total_sales bigint		
1	2022		1	Automotive			420473		
2	2022		2	Automotive			431998		
3	2022		3	Automotive			514582		
4	2022		4	Automotive			504116		
5	2022		5	Automotive			483482		
6	2022		6	Automotive			484120		
7	2022		7	Automotive			466716		
8	2022		8	Automotive			499292		
9	2022		9	Automotive			460062		
10	2022		10	Automotive			467212		
11	2022		11	Automotive			435198		
12	2022		12	Automotive			456984		
Total rows: 12 of 12			Query complete 00:00:00.098						

3)Top-performing industries in terms of sales for a year 2020, and how do their sales compare month-over-month?

WITH monthly_sales AS (

SELECT

year,

month,

industry,

SUM(sales) AS total_sales

FROM

retail_sales

WHERE

year = 2020

GROUP BY

```

        year,
        month,
        industry
    ),
    top_industries AS (
        SELECT
            year,
            month,
            industry,
            total_sales,
            RANK() OVER (PARTITION BY year, month ORDER BY total_sales DESC) AS
            industry_rank
        FROM
            monthly_sales
    )
    SELECT
        year,
        month,
        industry,
        total_sales
    FROM
        top_industries
    WHERE
        industry_rank = 1
    ORDER BY
        year,
        month;

```

Data Output		Messages		Notifications	
	year integer	month integer	industry text	total_sales bigint	
1	2020	1	Automotive	342994	
2	2020	2	Automotive	358432	
3	2020	3	Automotive	295174	
4	2020	4	Automotive	245248	
5	2020	5	Automotive	380186	
6	2020	6	Automotive	402878	
7	2020	7	Automotive	413128	
8	2020	8	Automotive	413998	
9	2020	9	Automotive	405586	
10	2020	10	Automotive	411928	
11	2020	11	Automotive	371776	
12	2020	12	Automotive	421624	
Total rows: 12 of 12		Query complete 00:00:00.104			

4)Top-performing industries in terms of sales for a year 2019, and how do their sales compare month-over-month?

```

WITH monthly_sales AS (
  SELECT
    year,
    month,
    industry,
    SUM(sales) AS total_sales
  FROM
    retail_sales
  WHERE
    year = 2019
  GROUP BY
    year,
    month,
    industry
),

```

```
top_industries AS (  
    SELECT  
        year,  
        month,  
        industry,  
        total_sales,  
        RANK() OVER (PARTITION BY year, month ORDER BY total_sales DESC) AS  
industry_rank  
    FROM  
        monthly_sales  
)  
SELECT  
    year,  
    month,  
    industry,  
    total_sales  
FROM  
    top_industries  
WHERE  
    industry_rank = 1  
ORDER BY  
    year,  
    month;
```

Data Output					Messages	Notifications
	year integer	month integer	industry text	total_sales bigint		
1	2019	1	Automotive	329164		
2	2019	2	Automotive	335628		
3	2019	3	Automotive	411614		
4	2019	4	Automotive	383028		
5	2019	5	Automotive	414218		
6	2019	6	Automotive	387080		
7	2019	7	Automotive	404848		
8	2019	8	Automotive	426178		
9	2019	9	Automotive	369572		
10	2019	10	Automotive	384366		
11	2019	11	Automotive	375498		
12	2019	12	Automotive	385392		
Total rows: 12 of 12					Query complete 00:00:00.150	

Business Question

5) Which specific kind of businesses contribute the most to total sales, and how does their performance vary across industries?

```

SELECT
    kind_of_business,
    industry,
    SUM(sales) AS total_sales
FROM
    retail_sales
GROUP BY
    kind_of_business,
    industry
ORDER BY
    total_sales DESC;

```


Data Output

Messages

Notifications

≡+

📄

▼

📋

▼

🗑️

🗄️

⬇️

📈

	kind_of_business text	industry text	total_sales bigint
1	Motor vehicle and parts dealers	Automotive	14531121
2	Automobile and other motor vehicle dealers	Automotive	13310756
3	Automobile dealers	Automotive	12389347
4	New car dealers	Automotive	10980583
5	Food and beverage stores	Food & Beverage	9451004
6	General merchandise stores	General Merchandise	9038491
7	Food services and drinking places	Restaurants & Bars	8635492
8	Nonstore retailers	Miscellaneous	8511856
9	Grocery stores	Food & Beverage	8463197
10	Supermarkets and other grocery (except convenience) sto...	Food & Beverage	8092220
11	Restaurants and other eating places	Restaurants & Bars	7541873

Total rows: 58 of 58

Query complete 00:00:00.261

6) Is there any seasonality in sales for specific industries, and how do they perform month-over-month?

```

SELECT
    industry,
    year,
    month,
    SUM(sales) AS total_sales
FROM
    retail_sales
GROUP BY
    year,
    industry,
    month
ORDER BY
    year,
    industry,
    month;

```

Data Output		Messages		Notifications	
≡ + 📄 ▼ 📋 ▼ 🗑		🗄		⬇ 📈	
	industry text 🔒	year integer 🔒	month integer 🔒	total_sales bigint 🔒	
1	Automotive	2010	1	184064	
2	Automotive	2010	2	190554	
3	Automotive	2010	3	247908	
4	Automotive	2010	4	234756	
5	Automotive	2010	5	238892	
6	Automotive	2010	6	235548	
7	Automotive	2010	7	245772	
8	Automotive	2010	8	242440	
9	Automotive	2010	9	226470	
10	Automotive	2010	10	225750	
11	Automotive	2010	11	216494	
Total rows: 1000 of 1872		Query complete 00:00:00.213			

7) How does the sales distribution vary among industries based on their North American Industry Classification System (NAICS) codes?

```

SELECT
    naics_code,
    industry,
    SUM(sales) AS total_sales
FROM
    retail_sales
GROUP BY
    naics_code,
    industry
ORDER BY
    naics_code,
    total_sales DESC;

```

Data Output Messages Notifications			
+ 📄 ▼ 📋 ▼ 🗑️ 🗄️ ⬇️ 📈			
	naics_code text	industry text	total_sales bigint
1	44,114,412	Automotive	13310756
2	441	Automotive	14531121
3	4411	Automotive	12389347
4	44111	Automotive	10980583
5	44112	Automotive	1408764
6	4413	Automotive	1220364
7	442	Home Goods & Electronics	1434266
8	442,443	Home Goods & Electronics	2674826
9	4421	Home Goods & Electronics	775943
10	4422	Home Goods & Electronics	514352
11	44221	Home Goods & Electronics	123189
Total rows: 58 of 58		Query complete 00:00:00.109	

8) Are there any outliers or significant changes in sales for specific industries during particular months or years?

```

SELECT
    industry,
    year,
    month,
    sales
FROM
    retail_sales
WHERE
    (industry, year, month) IN (
        SELECT
            industry,
            year,
            month
        FROM (
            SELECT
                industry,
                year,

```

```

        month,
        sales,
        LAG(sales) OVER (PARTITION BY industry ORDER BY year, month) AS
prev_sales,
        LEAD(sales) OVER (PARTITION BY industry ORDER BY year, month) AS
next_sales
    FROM
        retail_sales
    ) AS sales_analysis
    WHERE
        sales > 1.5 * COALESCE(prev_sales, 0) OR sales > 1.5 * COALESCE(next_sales, 0)
    )
ORDER BY
    industry,
    year,
    month;

```

9) Which businesses all-time average sale was above 10 billion dollars?

```

SELECT
    kind_of_business,
    AVG(sales) AS average_sale
FROM
    retail_sales
GROUP BY
    kind_of_business
HAVING
    AVG(sales) > 10000; -- 10 billion dollars in cents (1 dollar = 100 cents)

```

10) Which kind of businesses within the automotive industry had the highest sales revenue for 2022?

```

SELECT
    kind_of_business,
    SUM(sales) AS total_sales
FROM

```

```

    retail_sales
WHERE
    industry = 'Automotive' AND year = 2022
GROUP BY
    kind_of_business
ORDER BY
    total_sales DESC
;

```

11)What is the contribution percentage of each business in the automotive industry this year?

```

WITH automotive_sales AS (
    SELECT
        kind_of_business,
        SUM(sales) AS total_sales
    FROM
        retail_sales
    WHERE
        industry = 'Automotive' AND
        year = 2022
    GROUP BY
        kind_of_business
),
total_sales_automotive AS (
    SELECT
        SUM(sales) AS total_sales_automotive
    FROM
        retail_sales
    WHERE
        industry = 'Automotive' AND
        year = 2022
)
SELECT
    kind_of_business,

```

```

ROUND(((total_sales / total_sales_automotive.total_sales_automotive) * 100, 2) AS
contribution_percentage
FROM
    automotive_sales
CROSS JOIN
    total_sales_automotive;

```

--

```

with total_sales as(select year, industry, sum(sales) as sales_sum
from retail_sales
GROUP BY 1,2)

```

```

SELECT curr.industry, prev.year as previous_year, curr.year as current_year,
    (curr.sales_sum - prev.sales_sum) / prev.sales_sum * 100 as YoY

from total_sales as curr
join total_sales as prev
    on curr.year=prev.year+1 AND curr.industry=prev.industry
ORDER BY industry, curr.year DESC;

```

12)What are the year-over-year growth rates for each industry per year?

```

with total_sales as(select year, industry, sum(sales) as sales_sum
from retail_sales
GROUP BY 1,2)

```

```

SELECT curr.industry, prev.year as previous_year, curr.year as current_year,
    (curr.sales_sum - prev.sales_sum) / prev.sales_sum * 100 as YoY

from total_sales as curr
join total_sales as prev
    on curr.year=prev.year+1 AND curr.industry=prev.industry
ORDER BY industry, curr.year DESC;

```

--OR--

```
SELECT
    year,
    industry,
    (sales - LAG(sales) OVER (PARTITION BY industry ORDER BY year)) / LAG(sales)
OVER (PARTITION BY industry ORDER BY year) * 100 AS growth_rate
FROM
    retail_sales
ORDER BY
    industry, year;
```

13)What are the yearly total sales for women's clothing stores and men's clothing stores?

```
SELECT
    year,
    sum(CASE WHEN kind_of_business = 'Women's clothing stores' THEN sales ELSE 0
END) as women_sales,
    sum(CASE WHEN kind_of_business = 'Men's clothing stores' THEN sales ELSE 0 END)
as men_sales
FROM
    retail_sales
GROUP BY
    year;
```

14)What is the yearly ratio of total sales for women's clothing stores to total sales for men's clothing stores?

```
SELECT year, women_sales/men_sales as Women_to_Men_ratio
FROM (
    SELECT year,
        sum(CASE WHEN kind_of_business = 'Women's clothing stores' THEN sales ELSE 0
END) as women_sales,
```

```

        sum(CASE WHEN kind_of_business = 'Men's clothing stores' THEN sales ELSE 0 END)
as men_sales
    FROM retail_sales
    GROUP BY 1
) subquery;

```

15)What is the year-to-date total sale of each month for 2019, 2020, 2021, and 2022 for the women’s clothing stores?

```

SELECT
    rs.month,
    rs.year,
    rs.sales,
    (
        (
            SELECT SUM(sales)
            FROM retail_sales rs2
            WHERE rs2.year = rs.year
            AND rs2.month <= rs.month
            AND rs2.kind_of_business = 'Women's clothing stores'
        )
    ) AS ytd_sales
FROM
    retail_sales AS rs
WHERE
    rs.kind_of_business = 'Women's clothing stores'
    AND rs.year IN (2019, 2020, 2021, 2022);

```

16)What is the month-over-month growth rate of women’s clothing businesses in 2022?

-- Query 1

```

SELECT
    month,
    sales AS current_sales,

```



```

-- now we want the sales from 1 previous period
LAG(sales, 1) OVER (ORDER BY month) AS prev_sales
FROM
    retail_sales
WHERE
    kind_of_business = 'Women\'s clothing stores'
    AND year = 2022;

-- Query 2
SELECT
    month,
    sales AS current_sales,
    LAG(sales, 1) OVER (ORDER BY month) AS prev_sales,
    (sales - LAG(sales, 1) OVER (ORDER BY month)) / LAG(sales, 1) OVER (ORDER BY
month) * 100 AS growth_rate
FROM
    retail_sales
WHERE
    kind_of_business = 'Women\'s clothing stores'
    AND year = 2022;

```

Data Source: U.S. A. Government, Open available data

GitHub Repo: <https://github.com/Gurukrishn>

Author: © 2025 [Gurukrishna Vadlamudi](#)